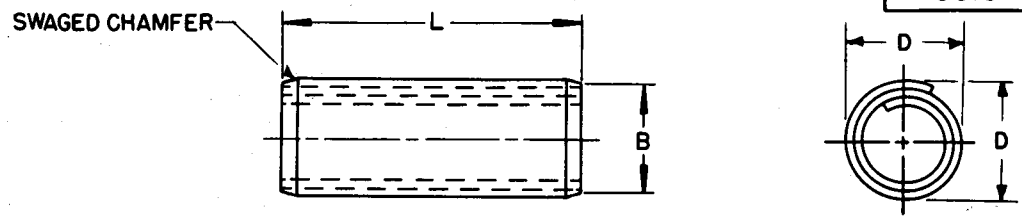
FED. SUP CLASS
5315



REVIEWER: AV, IS, MI, MU, NSA, 85
USER: AT, EL, GL, MC, ME, SH

TABLE I

NOMINAL DIAMETER	D DIA ACTUAL (FREE) STATE		B CHAMFER DIAMETER	MINIMUM DOUBLE SHEAR STRENGTH POUNDS		RECOMMENDED HOLE SIZE	
	MAX	MIN	MAX	CARBON STEEL 1070 THRU 1095	CRES 302	MAX	MIN
1/16	.073	.067	.059	170 *	135	.065	.061
5/64	.089	.083	.075	275 *	225	.081	.077
3/32	.106	.099	.091	375	300	.097	.093
7/64	.121	.114	.106	525	425	.112	.108
1/8	.139	.131	.121	675	550	.129	.124
5/32	.172	.163	.152	1100	875	.160	.155
3/16	.207	.196	.182	1500	1200	.192	.185
7/32	.240	.228	.214	2100	1700	.224	.217
1/4	.273	.260	.243	2700	2200	.256	.247
5/16	.339	.324	.304	4400	3500	.319	.308
* 3/8	.405	.388	.366	6000	5000	.383	.370
* 7/16	.471	.452	.427	8400	6700	.446	.431
* 1/2	.537	.516	.488	11000	8800	.510	.493

* Not manufactured

NOTES:

1. MATERIAL: Steel, Carbon, per Federal Standard No. 66, steel numbers 1070 thru 1095.
   Steel, Corrosion Resisting, per Federal Standard No. 66, steel number 302.
2. PROTECTIVE COATING: Steel, Carbon, cadmium plated per QQ-P-416, Type II, Class 3.
   Steel, Carbon, phosphate coated per MIL-P-16232, Type Z, Class 2.
   Steel, Corrosion Resisting, passivated per QQ-P-35.
3. HARDNESS: Steel, Carbon; Rockwell C46-53.
   Steel, Corrosion Resisting (302); work hardened.
4. EDGES: All sharp edges shall be broken.
5. DIMENSIONS: All dimensions are in inches.
6. TOLERANCES: Length L: All diameters thru 3/8":
   ±.010" for lengths thru 2"
   ±.015 for lengths over 2" thru 3"
   ±.025 for lengths over 3"
   All diameters larger than 3/8"
   ±.025 for all lengths
7. PART NUMBER: The MS part number consists of the MS number plus the dash number.
   Example: MS51987-13.
8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
9. Referenced documents shall be of the issue in effect on the date of invitations for bid.
10. For functional limitations on usage of spring pins, see MS33547.
11. Patent number 2737843 patented 13 March 1956. The Government has a royalty free license under this patent for the benefit of manufacturers of this item either for the Government or for use in equipment to be delivered to the Government.

(A) ENTIRE STANDARD REVISED

APPROVED 25 MARCH 1965 REVISED (A) 23 MAY 1969

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust 82	TITLE PIN, SPRING—TUBULAR, COILED, LIGHT DUTY	MILITARY STANDARD MS51987
PROCUREMENT SPECIFICATION MIL-P-10971	SUPERSEDES: BFSX3 and BFSX3.1	SHEET 1 OF 4

DD FORM 1 SEP 67 672-1 (Coordinated)

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TABLE II

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 302
	DASH NUMBER		
1/16 NOMINAL DIAMETER			
5/16			369
3/8			370
1/2			371
5/8			372
3/4			373
7/8			374
1			375
5/64 NOMINAL DIAMETER			
3/8			376
1/2			377
5/8			378
3/4			379
7/8			380
1			381
3/32 NOMINAL DIAMETER			
3/8	13	47	382
7/16	367	368	383
1/2	14	48	384
5/8	15	49	385
3/4	16	50	386
7/8	17	51	387
1	18	52	388
7/64 NOMINAL DIAMETER			
3/8	19	53	389
1/2	20	54	390
5/8	21	55	391
3/4	22	56	392
7/8	23	57	393
1	24	58	394
1-1/4	25	59	395
1-1/2	26	60	396

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 302
	DASH NUMBER		
1/8 NOMINAL DIAMETER			
3/8	27	61	397
1/2	28	62	398
5/8	29	63	399
3/4	30	64	400
7/8	31	65	401
1	32	66	402
1-1/4	33	67	403
1-1/2	34	68	404
5/32 NOMINAL DIAMETER			
1/2	103	147	405
5/8	104	148	406
3/4	105	149	407
7/8	106	150	408
1	107	151	409
1-1/4	108	152	410
1-1/2	109	153	411
1-3/4	110	154	412
2	111	155	413
3/16 NOMINAL DIAMETER			
1/2	112	156	414
5/8	113	157	415
3/4	114	158	416
7/8	115	159	417
1	116	160	418
1-1/4	117	161	419
1-1/2	118	162	420
1-3/4	119	163	421
2	120	164	422
2-1/4	121	165	423
2-1/2	122	166	424

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 302
	DASH NUMBER		
7/32 NOMINAL DIAMETER			
1/2	123	167	*
5/8	124	168	425
3/4	125	169	426
7/8	126	170	427
1	127	171	428
1-1/4	128	172	429
1-1/2	129	173	430
1-3/4	130	174	431
2	131	175	432
2-1/4	132	176	433
2-1/2	133	177	434
1/4 NOMINAL DIAMETER			
1/2	134	178	*
5/8	135	179	435
3/4	136	180	436
7/8	137	181	437
1	138	182	438
1-1/4	139	183	439
1-1/2	140	184	440
1-3/4	141	185	441
2	142	186	442
2-1/4	143	187	443
2-1/2	144	188	444
2-3/4	145	189	445
3	146	190	446
5/16 NOMINAL DIAMETER			
1	235	279	447
1-1/4	236	280	448
1-1/2	237	281	449
1-3/4	238	282	450
2	239	283	451
2-1/4	240	284	452
2-1/2	241	285	453
2-3/4	242	286	454
3	243	287	455

TABLE III. SUPERSEDED. The following are not manufactured or are not authorized for new design.

LENGTH L	* CARBON STEEL 1070 - 1095 CADMIUM	* CARBON STEEL 1070 - 1095 PHOSPHATE
	DASH NUMBER	
1/16 NOMINAL DIAMETER		
3/8	1	35
1/2	2	36
5/8	3	37
3/4	4	38
7/8	5	39
1	6	40

LENGTH L	* CARBON STEEL 1070 - 1095 CADMIUM	* CARBON STEEL 1070 - 1095 PHOSPHATE
	DASH NUMBER	
5/64 NOMINAL DIAMETER		
3/8	7	41
1/2	8	42
5/8	9	43
3/4	10	44
7/8	11	45
1	12	46

* Not manufactured

REVIEWER: AV, IS, MI, MU, NSA, 85
USER: AT, EL, GL, MC, ME, SH

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 25 MARCH 1965 REVISED (A) FOR CHANGES SEE SHEET 1

P.A. WC
Other Cust 82

TITLE
PIN, SPRING — TUBULAR, COILED, LIGHT DUTY

MILITARY STANDARD
MS51987
SHEET 2 OF

PROCUREMENT SPECIFICATION
MIL-P-10971

SUPERSEDES:
BFSX3 and BFSX3.1

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TABLE III. SUPERSEDED. The following are not manufactured or are not authorized for new design. (CONTINUED)

FED. SUP CLASS 5315

LENGTH L	** CRES 420 DASH NUMBERS									
	1/16 NOM DIA	5/64 NOM DIA	3/32 NOM DIA	7/64 NOM DIA	1/8 NOM DIA	5/32 NOM DIA	3/16 NOM DIA	7/32 NOM DIA	1/4 NOM DIA	5/16 NOM DIA
3/8	69	75	81	87	95					
1/2	70	76	82	88	96	191	200	211	222	
5/8	71	77	83	89	97	192	201	212	223	
3/4	72	78	84	90	98	193	202	213	224	
7/8	73	79	85	91	99	194	203	214	225	
1	74	80	86	92	100	195	204	215	226	323
1-1/4				93	101	196	205	216	227	324
1-1/2				94	102	197	206	217	228	325
1-3/4						198	207	218	229	326
2						199	208	219	230	327
2-1/4							209	220	231	328
2-1/2							210	221	232	329
2-3/4									233	330
3									234	331

REVIEWER: AV, IS, MI, MU, NSA, 85
USER: AT, EL, GL, MC, ME, SH

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 420
	DASH NUMBER		
* 3/8 NOMINAL DIAMETER			
1	244	288	332
1-1/4	245	289	333
1-1/2	246	290	334
1-3/4	247	291	335
2	248	292	336
2-1/4	249	293	337
2-1/2	250	294	338
2-3/4	251	295	339
3	252	296	340
3-1/4	253	297	341
3-1/2	254	298	342
3-3/4	255	299	343
4	256	300	344

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 420
	DASH NUMBER		
* 7/16 NOMINAL DIAMETER			
1-1/2	257	301	345
1-3/4	258	302	346
2	259	303	347
2-1/4	260	304	348
2-1/2	261	305	349
2-3/4	262	306	350
3	263	307	351
3-1/4	264	308	352
3-1/2	265	309	353
3-3/4	266	310	354
4	267	311	355

LENGTH L	CARBON STEEL 1070 - 1095 CADMIUM	CARBON STEEL 1070 - 1095 PHOS	CRES 420
	DASH NUMBER		
* 1/2 NOMINAL DIAMETER			
1-1/2	268	312	356
1-3/4	269	313	357
2	270	314	358
2-1/4	271	315	359
2-1/2	272	316	360
2-3/4	273	317	361
3	274	318	362
3-1/4	275	319	363
3-1/2	276	320	364
3-3/4	277	321	365
4	278	322	366

TABLE IV - INTERCHANGEABILITY

The following pins covered by dash numbers given in MS51987(WC) are inactivated or not manufactured after 23 MAY 1969 and are not to be used for new design. For new design and replacement, the following is recommended.

SUPERSEDED MS51987(WC) DASH NUMBER	RECOMMENDED REPLACEMENTS MS51987 DASH NUMBER
1	370
2	371
3	372
4	373
5	374
6	375
7	376
8	377
9	378
10	379
11	380
12	381
35	370
36	371
37	372
38	373
39	374
40	375
41	376
42	377

SUPERSEDED MS51987(WC) DASH NUMBER	RECOMMENDED REPLACEMENTS MS51987 DASH NUMBER
43	378
44	379
45	380
46	381
69	370
70	371
71	372
72	373
73	374
74	375
75	376
76	377
77	378
78	379
79	380
80	381
81	382
82	384
83	385
84	386

SUPERSEDED MS51987(WC) DASH NUMBER	RECOMMENDED REPLACEMENTS MS51987 DASH NUMBER
85	387
86	388
87	389
88	390
89	391
90	392
91	393
92	394
93	395
94	396
95	397
96	398
97	399
98	400
99	401
100	402
101	403
102	404
191	405
192	406

* Not manufactured
** Inactivated. Do not use for new design.

APPROVED 25 MARCH 1965 REVISED (A) FOR CHANGES SEE SHEET 1

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P.A. WC Other Cust 82	TITLE PIN, SPRING—TUBULAR, COILED, LIGHT DUTY	MILITARY STANDARD MS51987
PROCUREMENT SPECIFICATION MIL-P-10971	SUPERSEDES: BFSX3 and BFSX3.1	SHEET 3 OF

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FED. SUP CLASS 5315

REVIEWER: AV, IS, MI, MU, NSA, 85
USER: AT, EL, GL, MC, ME, SH

TABLE IV - INTERCHANGEABILITY (CONTINUED)

SUPERSEDED	RECOMMENDED REPLACEMENTS	SUPERSEDED	RECOMMENDED REPLACEMENTS	SUPERSEDED	RECOMMENDED REPLACEMENTS
MS51987(WC)	MS51987	MS51987(WC)	MS51987	MS51987(WC)	MS51987
DASH NUMBER		DASH NUMBER		DASH NUMBER	
193	407	254	*	315	*
194	408	255	*	316	*
195	409	256	*	317	*
196	410	257	*	318	*
197	411	258	*	319	*
198	412	259	*	320	*
199	413	260	*	321	*
200	414	261	*	322	*
201	415	262	*	323	447
202	416	263	*	324	448
203	417	264	*	325	449
204	418	265	*	326	450
205	419	266	*	327	451
206	420	267	*	328	452
207	421	268	*	329	453
208	422	269	*	330	454
209	423	270	*	331	455
210	424	271	*	332	*
211	*	272	*	333	*
212	425	273	*	334	*
213	426	274	*	335	*
214	427	275	*	336	*
215	428	276	*	337	*
216	429	277	*	338	*
217	430	278	*	339	*
218	431	288	*	340	*
219	432	289	*	341	*
220	433	290	*	342	*
221	434	291	*	343	*
222	*	292	*	344	*
223	435	293	*	345	*
224	436	294	*	346	*
225	437	295	*	347	*
226	438	296	*	348	*
227	439	297	*	349	*
228	440	298	*	350	*
229	441	299	*	351	*
230	442	300	*	352	*
231	443	301	*	353	*
232	444	302	*	354	*
233	445	303	*	355	*
234	446	304	*	356	*
244	*	305	*	357	*
245	*	306	*	358	*
246	*	307	*	359	*
247	*	308	*	360	*
248	*	309	*	361	*
249	*	310	*	362	*
250	*	311	*	363	*
251	*	312	*	364	*
252	*	313	*	365	*
253	*	314	*	366	*

* Not manufactured

APPROVED 25 MARCH 1965 REVISED (A) FOR CHANGES SEE SHEET 1

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P.A. WC Other Cust 82	TITLE PIN,SPRING-TUBULAR,COILED,LIGHT DUTY	MILITARY STANDARD MS51987
PROCUREMENT SPECIFICATION MIL-P-10971	SUPERSEDES: BFSX3 and BFSX3.1	SHEET 4 OF

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